Circles & Coordinate Geometry

Section 1: General Circle Equations & Completing the Square

1 A circle has the equation $x^2 + y^2 + 4x - 10y + 13 = 0$. What are its center and radius?

a. Center $(-2, 5)$, radius 4

b. Center $(2, -5)$, radius 4

c. Center $(-2, 5)$, radius 16

d. Center $(2, -5)$, radius 16

2 Find the equation of the circle that has points $A(1, 2)$ and $B(7, 10)$ as endpoints of a diameter.

a. $(x-4)^2 + (y-6)^2 = 25$

b. $(x-4)^2 + (y-6)^2 = 5$

c. $(x-3)^2 + (y-4)^2 = 25$

d. $(x+4)^2 + (y+6)^2 = 25$

3 For what value of k does $x^2 + y^2 - 6x + 8y + k = 0$ represent a single point (degenerate circle)?

a. 25

b. 16

c. 9

d. 0

Section 2: Tangent Lines to Circles

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4 What is the equation of the tangent line to the circle $x^2 + y^2 = 20$ at the point $(2, 4)$?

a. $x + 2y = 10$

b. $2x + y = 10$

c. $x - 2y = -6$

d. $2x - y = 0$

5 Find the length of the tangent line drawn from external point $P(8, 0)$ to the circle $x^2 + y^2 = 16$.

a. $4\sqrt{3}$

b. 4

c. 8

d. $2\sqrt{5}$

6 At what points does the line $y = x + 2$ intersect the circle $x^2 + y^2 = 10$?

a. $(1, 3)$ and $(-3, -1)$

b. $(3, 1)$ and $(-1, -3)$

c. $(1, 3)$ only

d. No intersection

Section 3: Loci & Conic Definitions

7 What is the locus of all points (x, y) that are equidistant from the point $(0, 3)$ and the line $y = -3$?

a. Parabola $y = \frac{1}{12}x^2$

b. Parabola $x = \frac{1}{12}y^2$

c. Circle $x^2 + y^2 = 9$

d. Ellipse $\frac{x^2}{9} + \frac{y^2}{6} = 1$

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8 What is the locus of all points whose distances to $(2, 0)$ and $(-2, 0)$ have a constant sum of 6?

a. Ellipse $\frac{x^2}{9} + \frac{y^2}{5} = 1$

b. Hyperbola $\frac{x^2}{9} - \frac{y^2}{5} = 1$

c. Circle $x^2 + y^2 = 9$

d. Parabola $y^2 = 4x$

9 What geometric figure is defined by $x = 4 + 3\cos t, y = -1 + 3\sin t$?

a. Circle centered at $(4, -1)$ with radius 3

b. Ellipse with semimajor axis 4

c. Hyperbola centered at $(4, -1)$

d. Line with slope 1

Section 4: Coordinate Proofs & Polygons

10 What type of quadrilateral has vertices at $A(0, 0), B(4, 0), C(5, 3),$ and $D(1, 3)$?

a. Parallelogram

b. Rhombus

c. Rectangle

d. Trapezoid (not parallelogram)

11 Find the centroid of the triangle with vertices $(1, 4), (5, -2),$ and $(6, 7)$.

a. $(4, 3)$

b. $(3, 4)$

c. $(4, 4)$

d. $(5, 3)$

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12 What is the slope of the altitude drawn to side BC in $\triangle ABC$ where $B = (2, 1)$ and $C = (6, 5)$?

a. -1

b. 1

c. $\frac{1}{2}$

d. -2

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